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Video Processing – HW3

Particle Filter Tracking – Part A

Course 0512.4263 2025/2026

Part A – Open Questions

Q1.

We saw that $\text{posterior} \propto \text{prior} \times \text{likelihood}$. Explain the formula and how it relates to the Kalman/Particle filter. Start from explaining what is the posterior, prior and likelihood.

Answer.

Q2.

For the measurement step in the particle filter, we used the histogram of the two patches (the patch around the new particle and the original patch of the object).

a. Why do we use the histogram to score the patches? what are the pros and cons of using this method?

Answer.

b. We took the patches and computed the histogram and their distances. Why didn't we use the SSD (sum square difference) between the patches (in the image space)?

Answer.

c. Can you think of another method to compare the patches? What are the pros and cons of the new method?

You can describe an example if it helps you.

Answer.

Q3.

Will particle filter work when the tracked object changes its scale (like a person walking towards and away from the camera)? What about the viewpoint of the object (like a person dancing)?

Answer.

Q4.

Can you think about a way to know when to update the template? Think of a scenario where we track a person. We keep a patch of the object's appearance in the first frame, and then compare each new patch to it. One of the problems is that, as time passes, the original patch no longer represents the object (light, viewpoint, ...). Can you think of a way to determine when to update the patch of the object? And how to determine the new patch (based on the previous frame/ based on the last two frames/...)?

Answer.